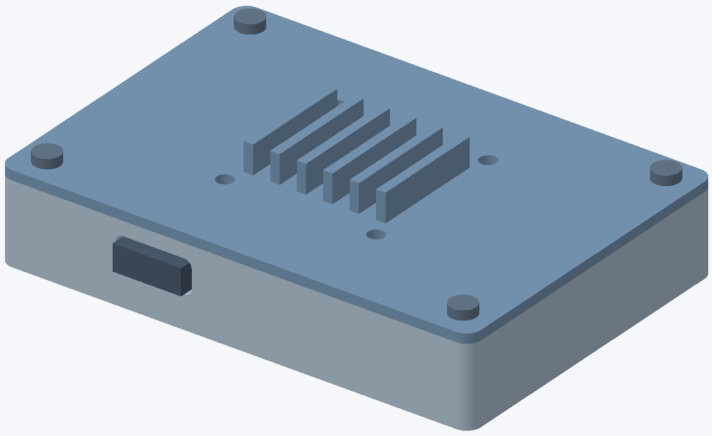
HARDWARE PROTOTYPING / PORTFOLIO

From packaging assumptions to a testable hardware design.

EDGE / 01 Compact edge-compute enclosure



EDGE / 01 | Rev B | CAD geometry; physical validation pending

100 × 70 × 25 mm	31 component pairs	8 W thermal study
Parametric passive housing	STEP + STL checked	Passive limit exposed

What I built. An independent generic compute enclosure with removable finned aluminum lid, integral thermal pedestal, CNC/print plans, dimensioned interfaces, an inspection fixture and rerunnable validation.

Engineering judgment. The assumed passive case reaches 93.4°C at 35°C ambient, missing an 80°C target. A fan hypothesis reaches 61.4°C but needs testing. The0.31–0.69 mm pad-gap stack makes supplier compression data and real board measurements the next design inputs.

Evidence. 31 component export pairs checked; 31 tests executed, 0 failures; 361 assembly pair checks.

Next. Print and inspect a fit article; qualify pad contact and fixture repeatability; machine supported metal parts; run a guarded heater test.

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Digital design and manufacturing planning. No completed fabrication or physical test is claimed.

github.com/gokulg846/compact-edge-enclosure